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18.03SC

Differential Equations 18.03SC — Differential Equations May 2026 **EXTREMELY****HARD – MIT LEVEL****Time Limit: 3 hours****Total Marks: 100**

1. (2 points) Prove the Hille-Yosida theorem: a linear operator A generates a strongly continuous contraction semigroup on a Banach space X iff A is closed, densely defined, and all $\lambda > 0$ are in the resolvent set with $\|R(\lambda, A)\| \leq 1/\lambda$.
2. (4 points) Compute the Fourier transform of the distribution $p.v.(1/x)$ and prove that $\widehat{p.v.(1/x)}(\xi) = -i\pi \operatorname{sgn}(\xi)$. Use this to solve the Hilbert transform equation $H(u) = f$.
3. (5 points) Classify the second-order linear PDE $u_{xx} + 2u_{xy} + u_{yy} + u_x + u_y = 0$. Determine its type (elliptic, parabolic, hyperbolic) and find its canonical form. Solve the resulting canonical equation.
4. (2 points) Construct a sequence of continuous functions $f_n : [0, 1] \rightarrow \mathbb{R}$ that converges pointwise to $f(x) = 0$ but does NOT converge uniformly. Prove your construction works and explain why uniform convergence fails.
5. (5 points) State and prove the Implicit Function Theorem. Apply it to show that the equation $F(x, y) = y^5 + y + x = 0$ can be solved for $y = g(x)$ near $(0, 0)$, and compute $g'(0)$.
6. (5 points) Prove the Poincaré inequality: for $u \in W_0^{1,p}(\Omega)$ with $\Omega \subset \mathbb{R}^n$ bounded, $\|u\|_{L^p} \leq C\|\nabla u\|_{L^p}$. Show that the constant C depends on the diameter of Ω and give the optimal constant for $p = 2$ on an interval.
7. (2 points) Prove the Nash embedding theorem for compact Riemannian manifolds: any compact C^∞ Riemannian manifold can be isometrically embedded into $\mathbb{R}^N$ for sufficiently large N .
8. (4 points) Prove the Rellich-Kondrachov compactness theorem: $W^{1,p}(\Omega) \hookrightarrow L^q(\Omega)$ compactly for $1 \leq q < p^*$ when Ω is bounded and $\partial\Omega$ is C^1 .
9. (2 points) Prove the Brouwer fixed-point theorem using differential topology: any continuous map $f : \overline{B}^n \rightarrow \overline{B}^n$ has a fixed point. Use the fact that there is no smooth retraction of the ball onto its boundary.

10. (2 points) Let (X, d) be a complete metric space and $T : X \rightarrow X$ a contraction mapping with contraction factor $k < 1$. Prove the Banach fixed-point theorem: T has a unique fixed point. Additionally, derive the error estimate $\|x_n - x^*\| \leq \frac{k^n}{1-k} \|x_1 - x_0\|$ where $x_{n+1} = T(x_n)$.
11. (3 points) Prove Sard's theorem: if $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ is C^k with $k > \max(n - m, 0)$, then the set of critical values of f has Lebesgue measure zero in $\mathbb{R}^m$.
12. (5 points) Prove that symplectic integrators preserve the symplectic structure $\omega = \sum dq^i \wedge dp_i$ of Hamiltonian systems. Show that the Verlet (Störmer) method is symplectic and has energy error bounded over exponentially long times for integrable systems.
13. (5 points) Prove that the Laplace operator Δ on a compact Riemannian manifold has discrete spectrum $\lambda_1 \leq \lambda_2 \leq \dots \rightarrow \infty$ and each eigenvalue has finite multiplicity. Show $\lambda_1 = \inf \frac{\|\nabla u\|^2}{\|u\|^2}$ over nonzero $u \in H^1(M)$ with $\int u = 0$.
14. (5 points) Analyze the stability of the Runge-Kutta method of order 4 when applied to the stiff ODE $y' = \lambda y$ with $\lambda \in \mathbb{C}$, $\operatorname{Re}(\lambda) < 0$. Determine the region of absolute stability and discuss how the step size h must be chosen for stiff problems.
15. (4 points) Prove the Schauder fixed-point theorem using Schauder estimates and the Arzela-Ascoli theorem: a compact operator $T : X \rightarrow X$ on a Banach space has a fixed point if it maps a closed convex set into itself.
16. (4 points) Prove the Divergence theorem for smooth vector fields on compact manifolds with boundary and derive the Green's identities.
17. (6 points) Prove the Lax-Milgram theorem and use it to show existence and uniqueness of weak solutions to the Dirichlet problem $-\Delta u = f$ in Ω , $u|_{\partial\Omega} = 0$ for $f \in L^2(\Omega)$.
18. (4 points) Prove that $\mathbb{R}P^3$ is diffeomorphic to $SO(3)$ by constructing explicit maps and verifying they are smooth inverses. Use this to compute $\pi_1(SO(3))$.
19. (4 points) Prove that the Fourier transform is an isometry on $L^2(\mathbb{R}^n)$ (Plancherel's theorem) and extends uniquely from $L^1 \cap L^2$ to a unitary operator on L^2 .
20. (5 points) Let $f_n : [0, 1] \rightarrow \mathbb{R}$ be a sequence of functions of bounded variation with $\operatorname{Var}(f_n) \leq C$ uniformly and $f_n \rightarrow f$ pointwise. Prove that f is of bounded variation and $\operatorname{Var}(f) \leq C$. Give a counterexample showing that uniform bounded variation is necessary.
21. (4 points) Prove the maximum principle for elliptic PDEs: if $Lu = -\Delta u + c(x)u \geq 0$ in Ω with $c \geq 0$, then $\max_{\overline{\Omega}} u \leq \max_{\partial\Omega} u^+$. Show how this implies uniqueness for the Dirichlet problem.

22. (6 points) Prove the Gagliardo-Nirenberg-Sobolev inequality: for $1 \leq p < n$, there exists $C = C(n, p) > 0$ such that $\|u\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|\nabla u\|_{L^p(\mathbb{R}^n)}$ for all $u \in C_c^1(\mathbb{R}^n)$, where $p^* = np/(n - p)$.
23. (5 points) State and prove the spectral theorem for compact self-adjoint operators on a Hilbert space. Show that such an operator has a countable set of eigenvalues that accumulate only at zero, and the eigenvectors form an orthonormal basis.
24. (2 points) Construct an exhaustion function (proper smooth function bounded below) on any smooth manifold and use it to prove that every smooth manifold admits a complete Riemannian metric.
25. (5 points) Prove the Atiyah-Singer index theorem for a simple case: compute the index of the Dirac operator on S^2 and verify it equals the integral of the $\hat{A}$ -genus.

Solutions

Solution 1:

($\Rightarrow$): If A generates $T(t)$, show $R(\lambda, A) = \int_0^\infty e^{-\lambda t} T(t) dt$, so $\|R(\lambda, A)\| \leq \int_0^\infty e^{-\lambda t} \|T(t)\| dt \leq 1/\lambda$. ($\Leftarrow$): Define $A_n = nAR(n, A)$ (Yosida approximants) which are bounded generators of $T_n(t) = e^{tA_n}$. Show $T_n(t)$ converges strongly to a semigroup $T(t)$ and its generator is A .

Solution 2:

$\langle p.v.(1/x), \varphi \rangle = \langle p.v.(1/x), \widehat{\varphi} \rangle$. Write $p.v.(1/x) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{x} \chi_{|x| > \varepsilon}$. Its Fourier transform: $\int_{|x| > \varepsilon} \frac{e^{-2\pi i x \xi}}{x} dx = -i\pi \operatorname{sgn}(\xi)$ as $\varepsilon \rightarrow 0$. The Hilbert transform $H(u) = \frac{1}{\pi} p.v. \int \frac{u(t)}{x-t} dt$ has symbol $\widehat{H(u)}(\xi) = -i \operatorname{sgn}(\xi) \widehat{u}(\xi)$, so $H^{-1} = -H$.

Solution 3:

The characteristic equation: $a_{11}dy^2 - 2a_{12}dxdy + a_{22}dx^2 = 0$ gives $1 \cdot dy^2 - 2 \cdot 1 \cdot dxdy + 1 \cdot dx^2 = 0$, so $(dy - dx)^2 = 0$. Double root: parabolic type. Characteristic curves: $y - x = c$. Canonical coordinates: $\xi = y - x$, $\eta = x$. Then $u_{\eta\eta} + u_\eta = 0$. Solve: $u(\xi, \eta) = A(\xi)e^{-\eta} + B(\xi)$, i.e. $u(x, y) = A(y - x)e^{-x} + B(y - x)$.

Solution 4:

Let $f_n(x) = \max(0, 1 - n|x - 1/n|)$ (triangular spikes at $1/n$). For any x , for $n > 1/x$, $f_n(x) = 0$, so $f_n \rightarrow 0$ pointwise. But $\|f_n\|_\infty = 1$ for all n , so $\sup_{x \in [0,1]} |f_n(x) - 0| = 1 \not\rightarrow 0$, hence convergence is not uniform. This shows pointwise convergence does not imply uniform convergence.

Solution 5:

IFT: Let $F : \mathbb{R}^{m+n} \rightarrow \mathbb{R}^n$ be C^1 , $F(a, b) = 0$, and $\partial F / \partial y(a, b)$ invertible. Then there exists $g : U \rightarrow \mathbb{R}^n$ C^1 near a with $F(x, g(x)) = 0$ and $Dg(x) = -(\partial F / \partial y)^{-1} \partial F / \partial x$. For $F(x, y) = y^5 + y + x$, $\partial F / \partial y = 5y^4 + 1$, at $(0, 0)$ this is $1 \neq 0$, so IFT applies. $g'(0) = -(\partial F / \partial y)^{-1} \partial F / \partial x = -1/1 = -1$.

Solution 6:

For $u \in C_c^\infty(\Omega)$, write $u(x) = \int_{-\infty}^{x_1} \partial_1 u(t, x'') dt$. By Hölder, $|u(x)|^p \leq (\operatorname{diam} \Omega)^{p-1} \int |\partial_1 u|^p dt$. Integrate over Ω to get $\|u\|_p^p \leq (\operatorname{diam} \Omega)^p \|\nabla u\|_p^p$. On $(0, \pi)$, the optimal constant for $p = 2$ is $C = 1/\pi$, attained by $u(x) = \sin(x)$.

Solution 7:

Sketch: Use the implicit function theorem to construct a short immersion $f_0 : M \rightarrow \mathbb{R}^N$. The metric g on M can be expressed as $f_0^* g_{\mathbb{R}^N} + h$. Use the Nash-Moser iteration to solve the system of PDEs $\partial_i f \cdot \partial_j f = g_{ij}$. The iteration converges by tame estimates on the linearized operator $L_f(h) = df \cdot \nabla h$.

Solution 8:

Take a bounded sequence $\{u_n\}$ in $W^{1,p}$. By the Sobolev-Gagliardo-Nirenberg inequality, it's bounded in L^{p^*} . Use the Frechet-Kolmogorov criterion for compactness in L^q : show $\|u_n(\cdot + h) - u_n\|_{L^q} \rightarrow 0$ uniformly as $h \rightarrow 0$ using difference quotients and the $W^{1,p}$ bound. Extract a subsequence convergent in L^q by the Arzela-Ascoli type argument.

Solution 9:

Suppose f has no fixed point. Define $r(x)$ = intersection of ray from $f(x)$ through x with ∂B^n . This gives a smooth retraction (after smooth approximation). By Sard's theorem, regular values exist. For a regular value y , $r^{-1}(y)$ is a 1-manifold with boundary, one endpoint at y . The other endpoint would be in B^n , impossible since r maps interior to boundary. Contradiction.

Solution 10:

Pick any $x_0 \in X$, define $x_{n+1} = T(x_n)$. Then $d(x_{n+1}, x_n) \leq k^n d(x_1, x_0)$. By triangle inequality, $d(x_{n+m}, x_n) \leq \sum_{i=0}^{m-1} k^{n+i} d(x_1, x_0) \leq \frac{k^n}{1-k} d(x_1, x_0)$. So $\{x_n\}$ is Cauchy, converges to x^* . By continuity of T , $T(x^*) = x^*$. Uniqueness: if $Tx = x$, $Ty = y$, then $d(x, y) = d(Tx, Ty) \leq kd(x, y)$, so $d(x, y) = 0$. The error estimate follows from taking $m \rightarrow \infty$.

Solution 11:

Cover U by cubes. On each cube where $\|Df\| \leq C$ and $|\det Df| \leq \varepsilon$ on a critical point, use the Whitney extension to show the image can be covered by $O((\varepsilon/C)^m)$ balls of radius $C/\sqrt{n}$. Summing over cubes gives arbitrarily small total measure. The differentiability condition ensures the critical set has enough structure.

Solution 12:

A one-step method Φ_h is symplectic if $(\Phi_h)^*\omega = \omega$. For Verlet: $q_{n+1/2} = q_n + \frac{h}{2}M^{-1}p_n$, $p_{n+1} = p_n - h\nabla V(q_{n+1/2})$, $q_{n+1} = q_{n+1/2} + \frac{h}{2}M^{-1}p_{n+1}$. Direct computation shows the Jacobian satisfies $J^T\Omega J = \Omega$. By the backward error analysis, the numerical solution is the exact flow of a nearby Hamiltonian, giving $O(e^{-c/h})$ energy conservation over exponentially long times.

Solution 13:

The resolvent $(-\Delta + I)^{-1} : L^2 \rightarrow H^2$ is compact by Rellich-Kondrachov. The spectral theorem for compact self-adjoint operators gives discrete eigenvalues $\mu_n \rightarrow 0$, so $\lambda_n = 1/\mu_n - 1 \rightarrow \infty$. Finite multiplicity follows from finite-dimensional eigenspaces of compact operators. The variational characterization follows from the Rayleigh quotient. The condition $\int u = 0$ removes the trivial $\lambda_0 = 0$.

Solution 14:

The RK4 method applied to $y' = \lambda y$ gives $y_{n+1} = R(z)y_n$ where $z = h\lambda$ and $R(z) = 1 + z + z^2/2 + z^3/6 + z^4/24$. The stability region is $\{z \in \mathbb{C} : |R(z)| \leq 1\}$. For stiff problems where $\operatorname{Re}(\lambda)$ is large negative, $|R(z)|$ grows for large $|z|$ (RK4 is not A-stable), requiring h small enough that $h\lambda$ stays in the stability region.

Solution 15:

Let $C \subset X$ be closed convex, $T(C) \subset C$, T compact. Approximate T by finite-rank operators T_n using Schauder basis. By Brouwer, each T_n has a fixed point x_n . By compactness of T , $\{T(x_n)\}$ has convergent subsequence $T(x_{n_k}) \rightarrow y$. Since $x_{n_k} = T_{n_k}(x_{n_k})$ and $T_{n_k} \rightarrow T$ uniformly, y is a fixed point of T .

Solution 16:

For $F \in \mathfrak{X}(\Omega)$, choose a partition of unity subordinated to a cover by coordinate

charts that are half-balls near boundary. In each chart, the divergence theorem reduces to integrating a partial derivative over a cube (or half-cube), which by Fubini and FTC gives boundary integrals. Summing yields $\int_{\Omega} \operatorname{div} F \, dV = \int_{\partial\Omega} F \cdot n \, dS$. Green's identities follow by setting $F = u \nabla v$.

Solution 17:

Lax-Milgram: Let $a : H \times H \rightarrow \mathbb{R}$ be coercive bilinear continuous on Hilbert H , $f \in H^*$. Then $\exists! u \in H$: $a(u, v) = f(v)$ for all $v \in H$. For the Dirichlet problem, $H = H_0^1(\Omega)$, $a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v$, $f(v) = \int_{\Omega} f v$. Poincaré inequality gives coercivity, trace theorem gives boundary values.

Solution 18:

Map $\mathbb{R}P^3 \rightarrow SO(3)$: represent a point in $\mathbb{R}P^3$ as a unit quaternion q modulo sign. The conjugation map $v \mapsto qvq^{-1}$ on pure quaternions gives a rotation in $SO(3)$. This is a smooth double cover $S^3 \rightarrow SO(3)$ with kernel $\{\pm 1\}$, inducing $\mathbb{R}P^3 \cong SO(3)$. Thus $\pi_1(SO(3)) \cong \mathbb{Z}/2\mathbb{Z}$.

Solution 19:

For $f \in L^1 \cap L^2$, define $\widehat{f}(\xi) = \int f(x) e^{-2\pi i x \cdot \xi} dx$. Show $\int |\widehat{f}|^2 = \int |f|^2$ by proving $\int \widehat{f} \widehat{g} = \int f \bar{g}$ for g Schwartz (using Gaussian approximation and Fubini). Extend by density of $L^1 \cap L^2$ in L^2 . The inverse Fourier transform gives the adjoint, and $\mathcal{F}^* = \mathcal{F}^{-1}$.

Solution 20:

For any partition $0 = x_0 < \dots < x_k = 1$, $|f(x_i) - f(x_{i-1})| = \lim_n |f_n(x_i) - f_n(x_{i-1})| \leq \liminf_n \operatorname{Var}(f_n) \leq C$. So $\operatorname{Var}(f) \leq C$. Counterexample without UB: $f_n(x) = nx$ on $[0, 1/n]$, $f_n(x) = 1$ elsewhere. $\operatorname{Var}(f_n) = 2n$, $f_n \rightarrow 1$ pointwise, but limit has variation 0, so inequality fails direction.

Solution 21:

Suppose u attains an interior maximum at x_0 with $u(x_0) = M > 0$ and $M > \max_{\partial\Omega} u^+$. At x_0 , $\Delta u \leq 0$ (Hessian negative semidefinite), so $Lu = -\Delta u + cu \geq 0 + cM \geq 0$. Strictly, if $c > 0$ or $M > 0$, this gives $Lu > 0$ at the maximum, contradiction. For uniqueness of $\Delta u = f$ with $u|_{\partial\Omega} = g$, difference of two solutions satisfies $\Delta w = 0$, $w|_{\partial\Omega} = 0$, so $w \equiv 0$.

Solution 22:

For $p = 1$, use $|u(x)| \leq \int_{-\infty}^{x_i} |\partial_i u| dx_i$, multiply over $i = 1, \dots, n$, integrate, apply Hölder repeatedly to get $\|u\|_{n/(n-1)} \leq \prod (\int |\partial_i u|)^{1/n} \leq \frac{1}{n} \sum \int |\partial_i u|$. For general p , apply the $p = 1$ case to $|u|^q$ with suitable q and use Hölder.

Solution 23:

Let $T : H \rightarrow H$ be compact self-adjoint. Show $\|T\| = \sup_{|x|=1} |\langle Tx, x \rangle|$ (achieved). Let $\lambda_1 = \pm \|T\|$ be the extremal eigenvalue with eigenvector e_1 . Inductively apply the same argument to T restricted to $\{e_1, \dots, e_{n-1}\}^\perp$. Compactness implies $\lambda_n \rightarrow 0$. The eigenvectors form an ONB by the spectral decomposition $Tx = \sum \lambda_n \langle x, e_n \rangle e_n$.

Solution 24:

Let $\{U_i\}$ be a countable cover by coordinate balls, $\{\rho_i\}$ a partition of unity. Define

$f = \sum_i i\rho_i$. This is proper: $f^{-1}([-\infty, c])$ is compact (only finitely many $i \leq c$ contribute). For a Riemannian metric, start with any metric g_0 , let $g = g_0 + df \otimes df$ on $f^{-1}((-\infty, \infty))$. The gradient of f has unit length, so any curve must escape to infinity in infinite length.

Solution 25:

The Dirac operator D on S^2 acts on spinors. $\text{ind}(D) = \dim \ker D - \dim \ker D^*$. By Lichnerowicz formula, $D^2 = \nabla^* \nabla + \frac{1}{4}R$. On S^2 with $R = 2$, $\ker D$ consists of the constant spinor. So $\text{ind}(D) = 1$. The $\hat{A}$ -genus: $\int_{S^2} \hat{A}(TS^2) = \int_{S^2} 1 - \frac{p_1}{24} + \dots = 1$ since $p_1(TS^2) = 2\chi = 4$ but the integral gives 0 at this order. Indeed $\int_{S^2} \frac{R}{4\pi} = \chi/2 = 1$.